

Introduction to CSS

(Cascading Style Sheets)



The Power of CSS



What is CSS?

Allow users to apply typographic and design styles for elements on a page

Done by using established HTML elements and specifying a rule for how that element should be displayed

Creates rules that specify how the content of an element should appear

Once you have learned how to write a CSS rule, learning CSS mostly involves  learning the different properties you can use

[W3Schools](https://www.w3schools.com/css/) is an excellent resource for this

Advantages

Greater page layout and style controls

Size, color, line spacing, placement, margins, etc.

Separates style from structure in the HTML document

Easier site maintenance

Make style changes to many pages (entire site) at once

Disadvantages

CSS not uniformly implemented in browsers

Older browsers don't support. Newer browsers have variances in display

Different versions of CSS are implemented (CSS2 vs CSS3)

HTML5 became standard on October 28, 2014 (hence all the reworking of lecture notes)

Versions

CSS 1 (1996) - No longer maintained

CSS 2 (1998) - No longer maintained

CSS 2.1 (2011) - Planned as the first & final revision of level 2, but work on CSS 2.2 began in 2015. This is the current "official" standard

CSS 3 (2012 - 2021) - Things got a lot more complicated here. CSS 3 is broken down into several distinct "modules," which are in varying stages of development. Still hasn't been "officially" adopted as the standard, though modern browsers mostly support it. (* **big** topic in CSCI 1720)

CSS 4 (?) - Proposed updates to CSS 3. Again, split into

The Power of CSS

HTML defines **structure**

CSS defines **style**

Separating the two concerns makes each easier to write and maintain

Back in the day, web designers had to use

```
<h1>  
  <font color="#3366ff" face="Helvetica">  
    Top of Page  
  </font>  
</h1>
```

...and would have to be applied to each `<h1>` in the document

The Power of CSS

HTML defines **structure**

CSS defines **style**

Separating the two concerns makes each easier to write and maintain

Is now

```
h1 {  
  color:  #3366ff;  
  font-family: "Helvetica";  
}
```

...and can now be applied to all `<h1>` elements at a time

Ways to Define Styles

One or more rules

contained in an external style sheet (linked)

contained in an internal style sheet (embedded)

applied directly to a tag (inline)

Ways to Define Styles

Style sheet rule syntax

Property/value pairs
separated by
semicolon

```
/* syntax */
selector {
    property: value;
}
/* examples */
body {
    margin: 0;
    font-size: 20px;
    line-height: 1.2
}
div {
    margin: 50px auto;
    width: 800px;
    padding: 50px;
}
p {
    width: 300px;
}
```

External Style Sheets

One way we can implement CSS is through **external (linked)** style sheets

With external style sheets we create a second document and name it with a **.css** extension, e.g., `style.css`

This second file contains only the CSS expressions

Preferred method for employing CSS

External stylesheets allow for modifying style across multiple HTML documents from a single location

External Style Sheets

Within every page that needs to use the external style sheet, we add the following tag to the head section of the document:

```
<link rel="stylesheet" href="css/style.css">
```

The advantage to external style sheets is that we can create one (or several) css file(s) and our entire website can use it/them

External Style Sheets

```
/* more examples */
p {
    width: 300px;
}
table {
    width: 300px;
    border-collapse: collapse;
}
td {
    padding: 4px;
    border: .5px solid #999;
    text-align: center;
}
h1 {
    color: #3366ff;
    font-family: "Helvetica";
}
```

```
/* more examples */
a {
    display: block;
    width: 100px;
    margin-right: 10px;
    text-decoration: none;
    text-align: center;
    background-color: #f9f9f9;
    color: #aa00aa;
    font-size: 16px;
    font-weight: bold;
    line-height: 1.5;
}
```

Embedded Style Sheets

Embedded style sheets are defined in the `<head>` section of the document

Only apply to the page in which they're included

Embedded style sheets are used to apply style to a single web page

```
<style>
  body {
    margin: 0;
    font-size: 20px;
    line-height: 1.2
  }
  div {
    margin: 50px auto;
    width: 800px;
    padding: 50px;
  }
  p {
    width: 300px;
  }
  h1 {
    color: #3366ff;
    font-family: "Helvetica";
  }
</style>
```

Embedded Style Sheets

```
<head>
  <meta charset="utf-8">
  <title>Title </title>

  <style>
    h1 {                                /* heading 1 */
      color:  red;      /* text color */
      font-size: 36px;                 /* text size */
    }
    p {                                /* paragraph */
      color:  green;    /* text color */
      font-family: Arial;              /* font family */
    }
  </style>
</head>
<body>
  <h1>This would display red text, 36px in size</h1>
  <p> This would display green text in Arial font</p>
</body>
```

Embedded Style Sheets

This would display red text, 36px in size

This would display green text in Arial font

Inline Style

Option three is declaring CSS inline

Inline allows us to add CSS to one specific tag within a document

```
<!-- Syntax
    <tag style="property: value; property: value">
-->
<h1 style="color:  red; font-size: 36px;">
    This would display red text, 36px in size
</h1>
<p style="color:  green; font-family: Arial;">
    This would display green text in Arial font"
</p>
```

Inline Style

This would display red text, 36px in size

This would display green text in Arial font

Order of Operations

Order of Operation

What happens if more than one style is applied to the same property for a tag?

The more specific rule is applied

In order of highest priority to lowest priority, we would have:

- Inline style

- Embedded style

- Linked Style

- Browser default

```

<head>
  <meta charset="utf-8">
  <title>Title</title>
  <style>
    h1 {                                /* heading 1 */
      color: green;                    /* text color */
      font-size: 36px;                /* text size */
    }
    em {                                /* em element */
      color: blue;                     /* text color */
    }
    h2 {                                /* heading 2 */
      color: red;                      /* text color */
      font-size: 36px;                /* font size */
    }
    h2 {                                /* heading 2 */
      font-size: 24px;                /* font size */
    }
  </style>
</head>
<body>
<h1><em>How does this line display and why?</em></h1>
<h2>How does this line display and why?</h2>
<h1 style="color: red;">
  How does this line display and why?
</h2>
</body>

```

How does this line display and why?

How does this line display and why?

How does this line display and why?

Order of Operation

What happens if more than one property's value is applied to an element using different forms of CSS (External, Embedded, or Inline)?

The more recent the definition has the higher priority (i.e., the rule that is **closest** to the element in the HTML that it affects)

Let's say in the below example, `style.css` has a rule that changes all paragraphs to a background color of blue

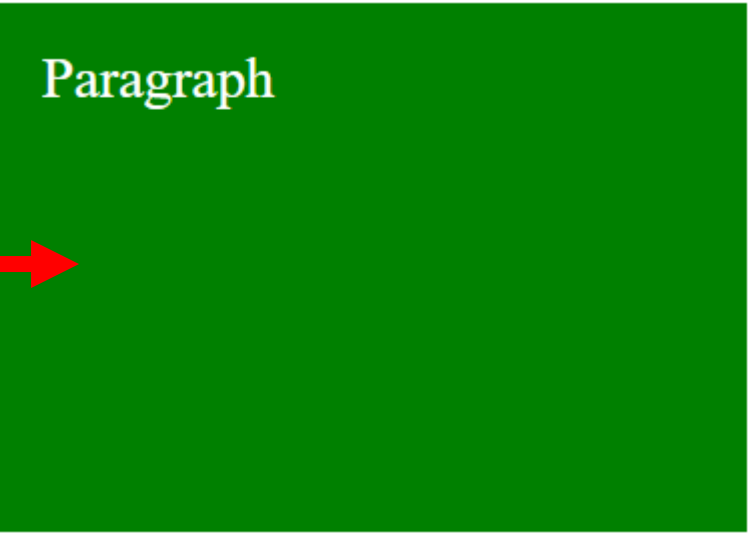
Order of Operation

```
p {  
  background-color: blue;  
  width: 300px;  
  height: 200px;  
  padding: 20px;  
  color: white;  
  font-size: 26px;  
}
```

style.css

```
<head>  
  <meta charset="utf-8">  
  <title>Title</title>  
  <link rel="stylesheet" href="style.css">  
  <style>  
    p {  
      background-color: green;  
      width: 300px;  
      height: 200px;  
      padding: 20px;  
    }  
  </style>  
</head>  
<body>  
  <p>  
    Paragraph  
  </p>  
</body>
```

index.html



Paragraph

Order of Operation

```
p {  
  background-color: blue;  
  width: 300px;  
  height: 200px;  
  padding: 20px;  
  color: white;  
  font-size: 26px;  
}
```

style.css

```
<head>  
  <meta charset="utf-8">  
  <title>Title</title>  
  <style>  
    p {  
      background-color: green;  
      width: 300px;  
      height: 200px;  
      padding: 20px;  
    }  
  </style>  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>  
  <p>  
    Paragraph  
  </p>  
</body>
```

index.html

Paragraph

Commenting

Commenting

In CSS, we can apply comments to embedded and linked CSS

To place a comment we use the syntax -- `/* COMMENT */`

```
/* paragraph elements */
p {
  background-color: blue;      /* background */
  width: 300px;               /* width */
  height: 200px;              /* height */
  padding: 20px;              /* padding */
  color: white;               /* color */
  font-size: 26px;            /* font-size */
}
```

Generally, you don't have to be this detailed with your comments (this is mostly for illustration)

But comments should be included in your code to explain what's going on

You might not be the person who has to update it later!

Commenting

For all .css files created, you must have the following comment block:

```
/*  
    Name:          Your Name  
    Course:        CSCI 1210  
    Assignment:    Lab x  
    Date:          xx.xx.2021  
    Description:    The purpose of this .css is to style  
                   the page(s) in this lab  
*/
```


IDs & Classes

ID Attribute

Every element on a web page can be assigned a **unique** ID so that it can be identified by other applications such as CSS or Javascript

A given ID can appear **only once** on a given page

Value should start with a letter or an underscore (_), not a number or any other

```
<!-- Syntax:
|   <tag id="id-name">
-->
<p id="demo">
|   This is a demonstration
</p>
```

ID Attribute

Since an ID is unique to a page, this is how we usually reference it in CSS

The octothorpe ("#") will apply the CSS rules to that element only

The second example is valid, but it isn't necessary to include the element name (see above)

```
/* creating an ID in
   CSS */
#md {
    width: 300px;
    font-size: 20px;
}

/* or */

p#md {
    width: 300px;
    font-size: 20px;
}
```

Class Attribute

Every element on a web page can also have one or more class attributes

Classes can be assigned to **multiple elements** on a page

Value **must** start with a letter or an underscore (_), not a number or any other character

Can assign multiple classes to a single element
space

```
<!-- Syntax:
    <tag class="class-name">
-->
<p class="red-text">
    This is a demonstration
</p>
```

```
<!-- Syntax:
    <tag class="class-name">
-->
<p class="red-text gray-background">
    This is a demonstration
</p>
```

Class Attribute

Again, classes can be used to modify an element's presentation, in multiple elements on a given page. These are the building-blocks used by CSS frameworks

To create a class, we use a dot (".")

```
/* creating a class in  
CSS */  
.md {  
    width: 300px;  
    font-size: 20px;  
}  
  
/* or */  
p.md {  
    width: 300px;  
    font-size: 20px;  
}
```

Class Attribute

Will be applied to any element that includes this class

More generic

```
/* creating a class in  
CSS */  
  
.md {  
    width: 300px;  
    font-size: 20px;  
}  
  
/* or */  
  
p.md {  
    width: 300px;  
    font-size: 20px;  
}
```


Class Attribute

Will be applied only to
paragraph elements

More specific

```
/* creating a class in  
CSS */  
.md {  
    width: 300px;  
    font-size: 20px;  
}  
  
/* or */  
  
p.md {  
    width: 300px;  
    font-size: 20px;  
}
```

CSS Text Properties

CSS Text Properties

`text-align:` (left, right, center, justified) -- `p{text-align:center;}`

`text-indent:` (pixels or percentage) -- `p{text-indent: 5%;}`

`text-decoration:` (none, overline, underline, line-through) -- `p{text-decoration: underline;}`

`word-spacing:` (pixels) -- `p{word-spacing: 5px;}`

`letter-spacing:` (pixels) -- `p{letter-spacing: 5px;}`

CSS Text Properties

`line-height:` (pixels, percentage, number) --

`p{line-height: 2;} /* this is double space */`

`color:` (name, hex, rgb, rgba) -- `p{color: red;}`

`font-family:` (specific,to,general) --

`p{font-family:"Times New Roman", Georgia, Serif;}`

CSS Text Properties

`font-size: (px, pt, em, or %) -- p{font-size: 10px;}`

`font-weight: (lighter, normal, bold, bolder, or value 100-900) --
p{font-weight:bold;}`

`font-style: (italic, oblique) -- p{font-style:italic;}`

CSS Background

background-color: (name, hex, rgb, rgba) -- **p{background-color: green; }**

background-image: url(imageLocation relative or abs) -- **p{background-image: url("images/bg.jpg"); }**

background-repeat: (no-repeat, repeat-x, repeat-y, repeat) -- **p{background-repeat: no-repeat; }**

background-size: (width and height in px or percent, or the word "cover") -- **p{background-size: cover; }**

CSS Background

`background-position:` (left, right, center, top, bottom, center)

`background-attachment:` (fixed, scroll)

```
body{  
    background-image: url("images/bg.gif");  
    background-repeat: no-repeat;  
    background-size: 100% 100%;  
}
```

CSS Colors

Colors

Effective use of colors can make or break a site

CSS has a number of ways of defining color

- Color keywords ("red," "green," "blue," etc.)

- RGB (Red, Green, Blue)

- Hexadecimal

- RGBa (Red, Green, Blue, Alpha)

- HSL (Hue, Saturation, Luminosity)

- HSLa (Hue, Saturation, Luminosity, Alpha)

We'll look at the first four

Colors

Option 1 - named colors

One option is to use the named color keywords.

There are 17 standard colors

aqua, black, blue, fuchsia, gray, green, lime,
maroon, navy, olive, orange, purple, red, silver,
teal, white, and yellow



There are 128 additional colors

BlanchedAlmond, ForestGreen, HoneyDew, WhiteSmoke,
RebeccaPurple

Colors

#663399Becca

Purple was the favorite color of Rebecca Alison Meyer who passed away twelve hours into her sixth birthday from brain cancer. Rebecca was the daughter of prolific CSS standards pioneer Eric Meyer. Eric kept his online colleagues informed of the battle his daughter and family were waging through blog posts and brief updates on Twitter.

After hearing the awful news, designer/author Jeffrey Zeldman decided to do something and started a Twitter hashtag campaign. What started as a hashtag charity campaign evidently transformed into a much larger project. Given Eric's prolific work on CSS, it was proposed that the hex-value #663399, a shade of purple, be aliased to "beccapurple."

When informed of the initiative, Eric had one request if the standards body were to adopt the proposal: call it "rebeccapurple" instead. Eric writes that "Rebecca informed us that she was about to be a big girl of six years old, and Becca was her baby name. Once she turned six, she wanted everyone (not just me) to call her Rebecca, not Becca."

Colors

Option 2 is to signify the decimal value that represents the color.

The colors on the screen are comprised of some mixture of Red, Green, and Blue (RGB)

Each color (R,G,B) can have a value from 0 (no color applied) to 255 (full color intensity)

Colors

Express the RGB values in a comma delimited list

This would be accomplished by `rgb(RED, GREEN, BLUE);`

Example: color: `rgb(255, 0, 0);` */* This would be red */*

Example: color: `rgb(255, 255, 255);` */* This would be white */*

Note: Do Not place a space between the 'b' in rgb and the opening paranthesis ('(')

Colors

	R	G	B
Red	255	0	0
Green	0	255	0
Blue	0	0	255
White	255	255	255
Black	0	0	0

Colors

Option 3 – Hexadecimal

Base 10 (decimal)	Base 16 (hexadecimal)
0	0
10	A
16	10
255	FF

Hexadecimal expresses single digits numbers 0-9, A, B, C, D, E, F. These numbers would be equivalent to decimal numbers 0-15 (with F being equal to 15)

To represent the values 0-255, two hexadecimal digits are needed

Colors

Option 3 – hexadecimal

To convert from decimal to hex, divide decimal number by 16. Quotient (expressed in hex) is left digit, remainder (expressed in hex) is right digit

$$175_{10} = 175/16 = 10 \text{ r } 15 = \text{AF}_{16}$$

$$219_{10} = 219/16 = 13 \text{ r } 11 = \text{DB}_{16}$$

To convert from hex to decimal, convert left digit to decimal and multiply by 16, add to that the right digit converted to decimal

$$\text{BC}_{16} = 11*16 + 12 = 188_{10}$$

$$\text{AA}_{16} = 10*16 + 10 = 176_{10}$$

Or, just use a
Scientific calculator

Colors

Option 3 – hexadecimal

Black	#000000
White	#FFFFFF
Red	#FF0000
Green	#00FF00
Blue	#0000FF
BlanchedAlmond	#FFEB3D
DeepSkyBlue	#00BFFF
GreenYellow	#ADFF2F



Colors

Option 4 would be to express the RGBA values

Very similar to option 2, however this allows us to identify the opacity of the object by setting the Alpha parameter. This ranges from 0.0 (fully transparent) to 1.0 (fully opaque)

This is supported in IE9+, Firefox 3+, Chrome, Safari, Opera 10+

`rgba(RED, GREEN, BLUE, ALPHA);`

Example: `color: rgba(255, 0, 0, 0.3); /* red with opacity */`

Colors

So to change the font color of a paragraph to white we could do one of the following:

```
p {  
    color: ■ white;  
} /* OR */  
p {  
    color: ■ #ffffff;  
} /* OR */  
p {  
    color: ■ rgb(255,255,255);  
} /* OR */  
p {  
    color: ■ rgba(255,255,255,1);  
}
```

Questions?



Lecture Quiz

1. Which of the following is not a valid class or id name?

A. contentTwo

B. _content

C. Content2

D. \$content

Lecture Quiz

2. What color is this: #00FF00?

A. 

B. 

C. 

D. 

Lecture Quiz

3. Given the following code, what background color of the paragraph?

- A. 
- B. 
- C. 
- D. 

```
<head>
  <style>
    p {
      background-color: #663399;
      color: #fff;
      font-size: 26px;
      width: 100px;
      height: 33px;
      padding: 20px;
    }
  </style>
</head>
<body>
  <p style="background-color: #00a;">
    A.
  </p>
</body>
```

Lecture Quiz

4. What CSS property would we use to modify the display size of text?

A. `font-size`

B. `text-size`

C. `size-font`

D. `text-points`

Lecture Quiz

5. There is a strong relationship between user experience and layout?

A. True

B. False

Lecture Quiz

6. Which is the correct way to assign multiple classes to an element?

- A. `<p class="md high-contrast centered">`
- B. `<p class="md, high-contrast, centered">`
- C. `<p class="md + high-contrast + centered">`
- D. `<p class="md/high-contrast/centered">`

Lecture Quiz

7. Which of the following correctly shows how to create a class in CSS?

A. `.image { }`

B. `p.image { }`

C. `div.image { }`

D. All of the above are correct

Lecture Quiz

8. What is the current, “official” version of CSS?

- A. 1
- B. 2
- C. 2.1
- D. 3
- E. 4

Lecture Quiz

9. What color is this: `rgb(255, 0, 0)`?

A. 

B. 

C. 

D. 

Lecture Quiz

10. Given the following CSS rule, which is a selector?

- A. p
- B. { }
- C. width
- D. 300px
- E. ;

```
p {  
    background-color: blue;  
    width: 300px;  
    height: 200px;  
    padding: 20px;  
    color: white;  
    font-size: 26px;  
}
```


Sources

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